W-S-893c

W-S-893c
April 2, 1964

SUPERSEDING
Int. Fed. Spec. W-S-00893b (GSA-FSS)
May 3, 1963 and
Fed. Spec. W-S-893a
April 18, 1956

FEDERAL SPECIFICATION

SWITCH, TOGGLE, AND MOUNTING STRAP, (INTERCHANGEABLE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers interchangeable, toggle (tumbler*) flush switches and mounting straps. The switches are designed for mounting in/on flush switch boxes by means of a non-integral strap.

1.2 Classification.

1.2.1 *Toggle switches.*

1.2.1.1 *Types, classes, and ratings.* Toggle switches furnished under this specification shall be of the following types, ratings, and classes as specified:

TABLE I.—*Types and ampere ratings*

Type I (AC, DC, and "T" rated)		Type II (AC)		
125 volts	250 volts	120 volts	120/277 volts	277 volts
10	5	15	15	—
20	10	20	20	15
		30	30	20

Classes:

Class 1—Single pole
Class 2—Double pole
Class 3—Three way
Class 4—Four way

* Tumbler—For the purpose of this specification, tumbler and toggle are considered synonymous.

1.2.2 *Mounting straps.*

1.2.2.1 *Styles.* Mounting straps furnished under this specification shall be of the following styles, as specified (see 6.2).

Style M—For one, two, or three devices, horizontal openings.

Style S—For one device, vertical opening.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington D. C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services

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Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL-E-17555—Electronic and Electrical Equipment And Associated Repair Parts, Preparation For Delivery Of.

Military Standards:

MIL-STD-105—Sampling Procedures And Tables For Inspection By Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publication of the issue in effect on date of invitation for bids, forms a part of this specification.

Underwriters' Laboratories, Inc., Standard:

UL-20—Standard For Snap Switches.

(Copies may be obtained from Underwriters' Laboratories, Inc., Walt Whitman Road, Melville, N. Y. 11749; 207 E. Ohio Street, Chicago, Illinois, 60611; and 1655 Scott Blvd., Santa Clara, California 95050.)

3. REQUIREMENTS

3.1 Switches and straps furnished under this specification shall conform to the applicable requirements of Underwriters Laboratories, Inc. UL-20, Standard For Snap Switches.

3.2 Fire and casualty hazards.

3.2.1 Each bidder shall submit to the contracting agency proof that the material he proposes to supply under this specification conforms to the applicable requirements of UL-20-Snap Switch. The label or listing with reexamination of the Underwriters' Laboratories, Inc., will be accepted as evidence that the material conforms to this requirement.

3.2.2 In lieu of the label or listing with reexamination, the bidder may submit independent proof satisfactory to the contracting agency that his material conforms to the applicable requirements of the published standards, including methods of tests, of UL-20.

3.2.3 Compliance with the above preliminary requirements in regard to fire and causalty hazards does not absolve the bidder from complete compliance with the requirements of this specification in order to secure acceptance of material.

3.3 Workmanship. Workmanship shall be in accordance with good commercial practice for this type of equipment.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 *For inspection.* A representative random sample shall be taken as shown. Acceptance or rejection of the lot shall be based on the number of defective units found in the sample selected and inspected, as being equal to or less than the specified

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acceptance number or equal to or greater than the specified rejection number as follows:

Sampling for visual and dimensional inspection

Lot size	Sample size	Acceptance number	Rejection number
15 and under	3	0	1
16 to 40	7	0	1
41 to 110	15	1	2
111 to 300	35	2	3
301 to 500	50	3	4
500 and over	75	4	5

NOTE. The Government inspector may institute tightened or reduced inspection in accordance with MIL-STD-105, AQL 2.5 percent defective.

The foregoing is a single sample procedure. Alternate double or multiple sampling procedures or alternate acceptable quality levels for different classes of defects may be specified at the discretion of the contracting officer.

4.2.2 *For test.* Not less than six switches shall be taken at random from each lot and tested in accordance with 4.4.

4.3 Inspection. Samples of switches and mounting straps taken in accordance with 4.2.1, shall be visually inspected for dimensional requirements, finish, functionability, and quality of materials and workmanship.

4.4 Tests. Sample switches taken in accordance with 4.2.2 shall be tested in accordance with Underwriters' Laboratories, Inc. UL-20 Standard For Snap Switches.

5. PREPARATION FOR DELIVERY

(For civil agency procurement, the definitions and applications of the various levels of packaging and packing, shall be as specified in Fed. Std. No. 102.)

5.1 Preservation and packaging. The preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 *Level A.* Switches shall be preserved and packaged in accordance with level A requirements of MIL-E-17555.

5.1.2 *Level C.* Switches shall be packaged in accordance with manufacturers' standard practice.

5.2 Packing. The packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* Switches shall be packed in accordance with the level A requirements of MIL-E-17555.

5.2.2 *Level B.* Switches shall be packed in accordance with the level B requirements of MIL-E-17555.

5.2.3 *Level C.* Switches shall be packed in a manner which will insure arrival at destination in satisfactory condition and which will be acceptable to the carrier at lowest rates.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking required in the contract or purchase order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to any special marking required in the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Electrical switches, with their associated mounting straps, as covered by this specification, are intended for use in any electrical circuit within their voltage and current limitations.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, symbol, and date of this specification.

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(*b*) Type, class, and ampere rating of switch (1.2.1.1).
(*c*) Style of mounting strap (1.2.2.1).
(*d*) Color switch handle.
(*e*) Inspection responsibility, if other than specified (4.1).
(*f*) Level of preservation, packaging, and packing required (5.1 and 5.2).
(*g*) Special marking, if required (5.3).

6.3 Transportation description. Transportation description applicable to this item is:

Electric switches or parts thereof, not otherwise indexed by name.
Carload minimum weight 30,000 pounds.
Electric switches.
Truckload minimum weight 30,000 pounds.

MILITARY CUSTODIANS:

Army—MO (ERDL)

Navy—Docks

Air Force—AFLC (SGEE) (85)

Review interests:

Army—MO (ERDL)

Navy—Docks

Air Force—AFLC (SGEE) (85)

User interest:

Navy—ESO

Air Force—ASD (11)

DSA—DESC

Military Coordinating Activity: Docks

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